

Gelivi Sai Chaitanya

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Education

University of Oklahoma

Norman, OK

Master of Science in Computer Science GPA: 3.91/4.00

Aug 2024 – May 2026

Relevant Coursework: Advanced Algorithms, Database Systems, Machine Learning, Computer Networks, Software Engineering, Distributed Systems, Computer Vision

Vardhaman College of Engineering

Hyderabad, India

Bachelor of Technology in Computer Science (AI & ML) GPA: 9.66/10.00

Dec 2020 – May 2024

Professional Experience

MassMutual

Remote

Machine Learning Intern

Aug 2022 – Jan 2023

- Built and deployed ARIMA/SARIMA and LSTM time-series models on 3+ years of transactional data (100k+ records across timestamps, product categories, and geography), improving monthly ticket volume forecast accuracy by 20%.
- Engineered ETL pipelines for multi-dimensional data ingestion, cleaning, and feature extraction, cutting preprocessing time by 30% and enabling faster model iteration cycles.
- Delivered interactive Matplotlib dashboards and executive reports translating model outputs into actionable operational decisions for resource planning teams.

Technical Skills

Languages: Python, Java, C/C++, JavaScript, SQL, Dart, HTML5, CSS3

Frameworks & Libraries: React, Node.js, Express.js, Flutter, Ethers.js, TensorFlow, Keras, Scikit-learn, Pandas, NumPy

Databases: PostgreSQL, MySQL, MongoDB, Oracle, Cloud Firestore

Tools & Platforms: Git, Docker, Linux, Hardhat, Firebase, Postman, PyInstaller, Jupyter

Concepts: Data Structures & Algorithms, OOP, REST APIs, Smart Contracts, Cryptography, CI/CD

Projects

Postgres Native Vector Retrieval | C, Python, PostgreSQL, pgvector, HNSW, PL/pgSQL, Docker

- Authored a native PostgreSQL extension in C implementing HNSW-based approximate nearest neighbor search directly inside the database engine, eliminating the need for an external vector store such as Pinecone or Weaviate and reducing infrastructure complexity by removing a network hop from every query.
- Achieved sub-10ms semantic query latency on 100k+ embedding datasets with 768-dimensional vectors, delivering a 5–10x speedup over a baseline brute-force kNN scan while maintaining recall@10 above 0.95.
- Benchmarked retrieval performance across multiple HNSW index configurations (graph connectivity and search depth parameters) and containerized the full stack with Docker Compose, enabling one-command reproducibility and cutting evaluation setup time from hours to minutes.

Secure Multi-Client Chat Application | Python, TCP Sockets, AES-256, PBKDF2, Tkinter, PyInstaller

- Engineered a concurrent multi-client TCP chat server in Python supporting 10+ simultaneous client connections through a multi-threaded architecture, enabling real-time message broadcast across all active sessions with sub-second delivery latency.
- Implemented end-to-end security using AES-256 message encryption and PBKDF2-based key derivation (100,000 iterations) with automatic per-session key rotation, enforcing forward secrecy and ensuring that compromise of one session key does not expose past or future messages.
- Designed a responsive Tkinter desktop GUI and packaged the entire application as a cross-platform standalone executable via PyInstaller, reducing the deployment footprint to a single binary with zero runtime dependencies for non-technical end users.

Credit Card Fraud Detection | Python, Scikit-learn, Hidden Naive Bayes, Bayesian Belief Networks

- Implemented Hidden Naive Bayes and Bayesian Belief Network classifiers on a highly imbalanced real-world financial transaction dataset (fraud cases under 1% of total volume), applying stratified sampling and decision threshold tuning to maximize F1-score on the minority fraud class.
- Evaluated both models against a standard Naive Bayes baseline using precision, recall, F1-score, and ROC-AUC; HNB delivered higher precision on fraudulent transactions, directly reducing false negatives — the most costly error class in financial fraud detection.
- **Research published at IEEE I2CT 2024** (peer-reviewed, 200+ attendees); demonstrated practical applicability of probabilistic graphical models for anomaly detection in production-scale financial systems. DOI: 10.1109/I2CT61223.2024.10544328

Leadership & Honors

- **Vice President**, Competitive Coding Club, Vardhaman College of Engineering (2022–2024) — grew participation by 40%; recipient, Best Club Activities Award, IEEE Envision 2024.
- **Cultural Committee Member**, Indian Student Association (ISA), University of Oklahoma (2024–Present).